

GeoNVS: Geometry Grounded Video Diffusion for Novel View Synthesis

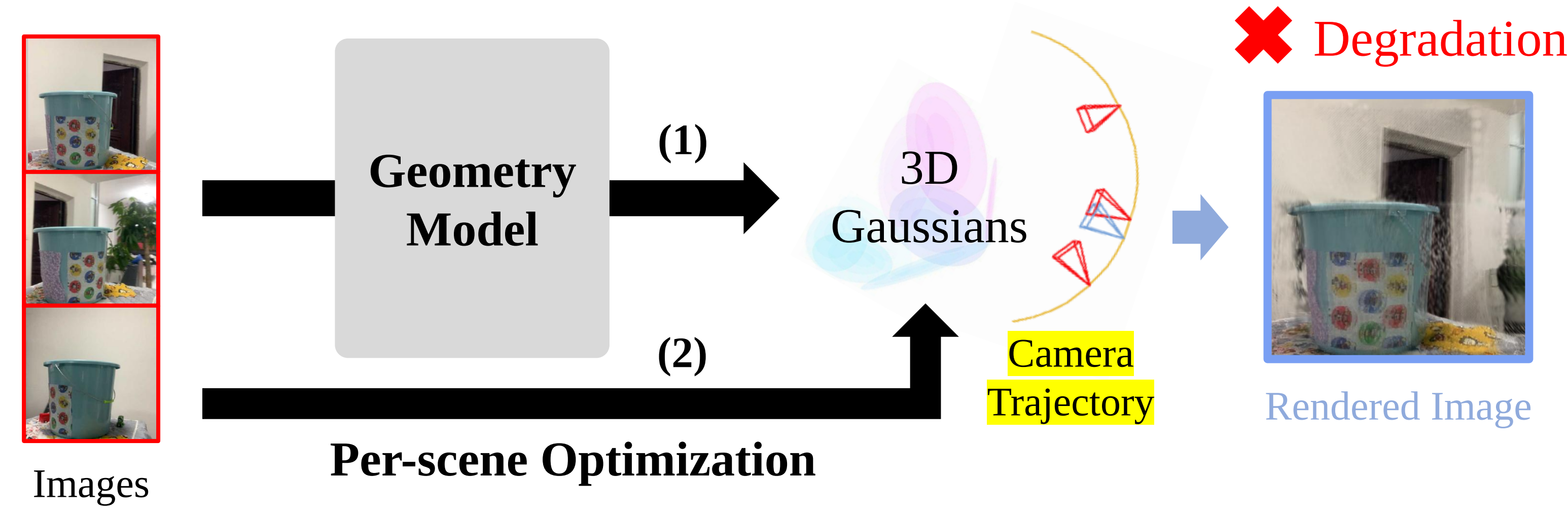
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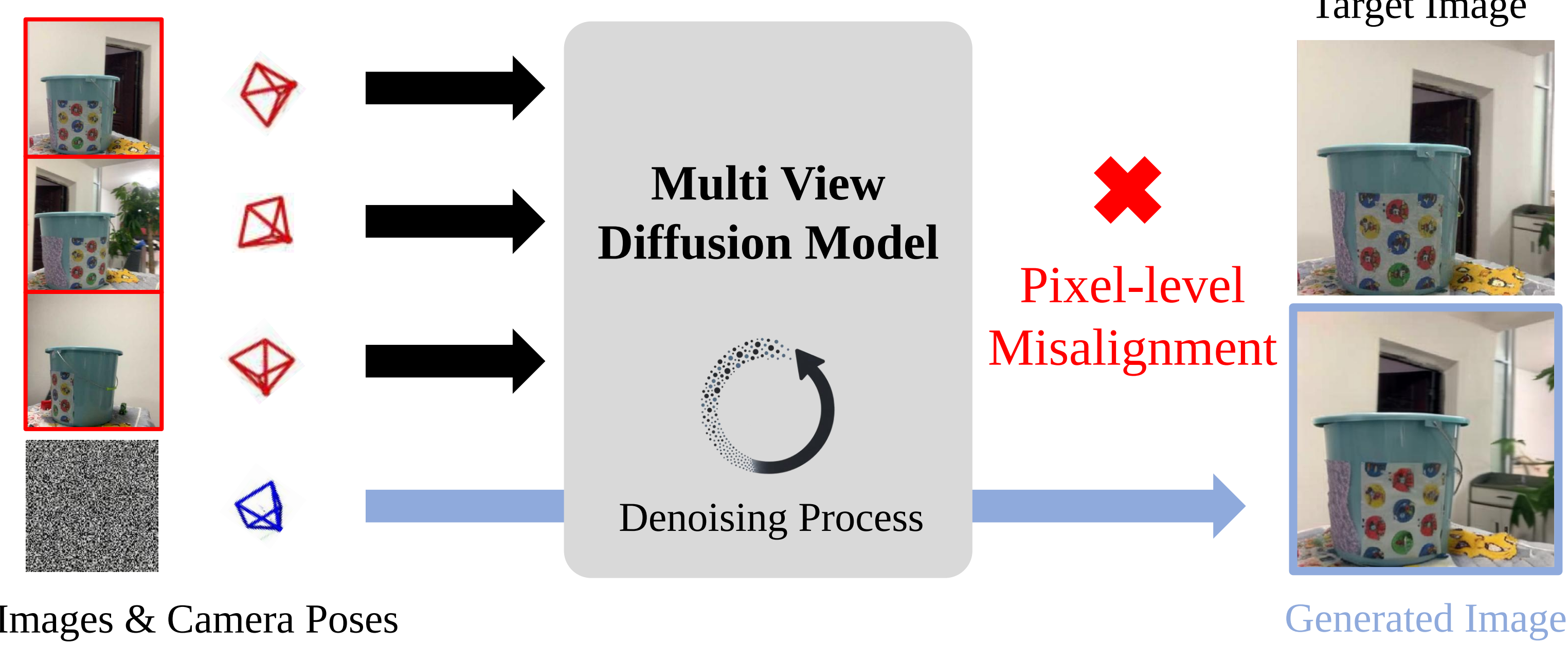
²Luma AI

Introduction

● Geometry-based Novel View Synthesis



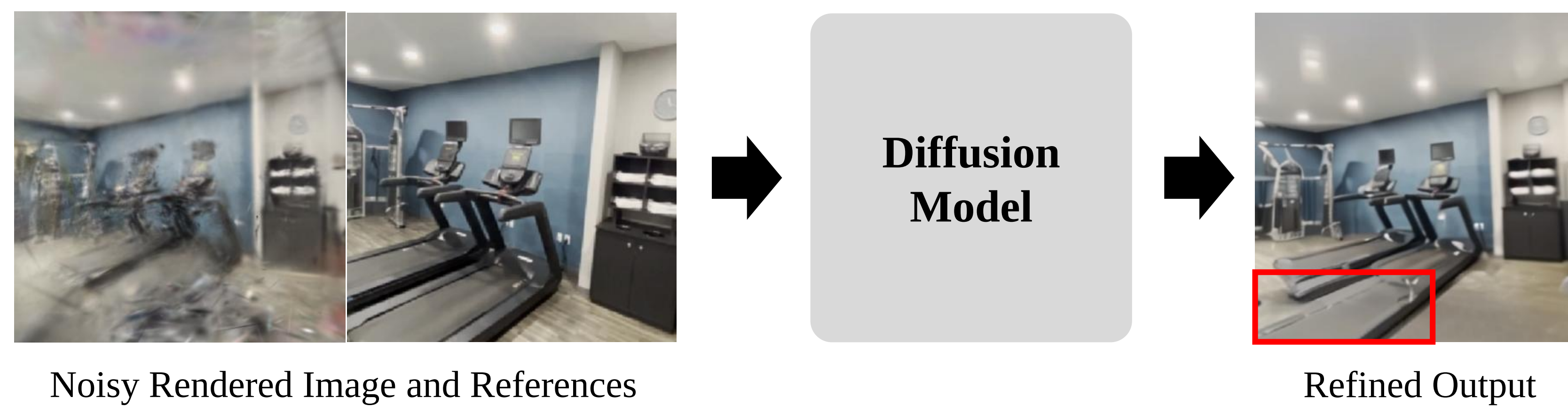
● Generative Novel View Synthesis



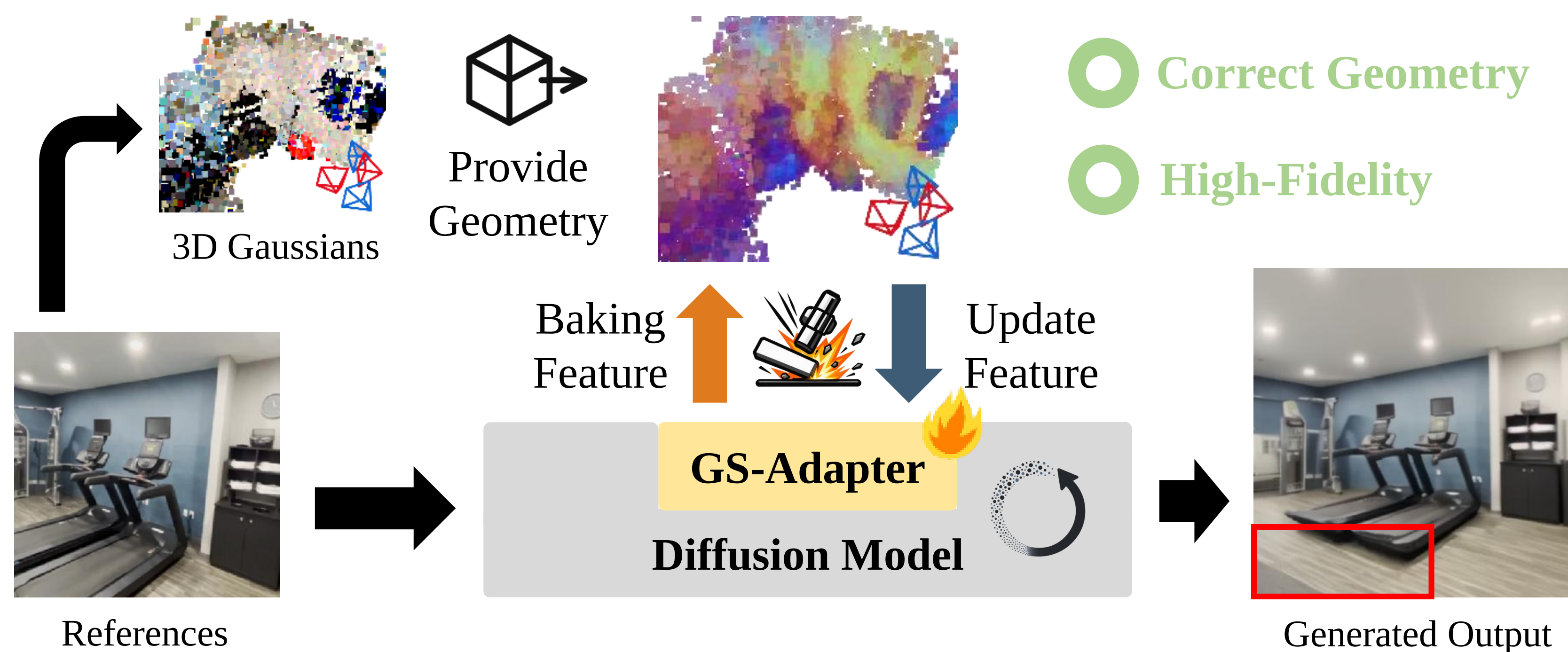
Motivation

● Baking Geometry into Video Generation Model

Previous Methods (Difix3D, MVSplat360 ...) : Render & Refinement

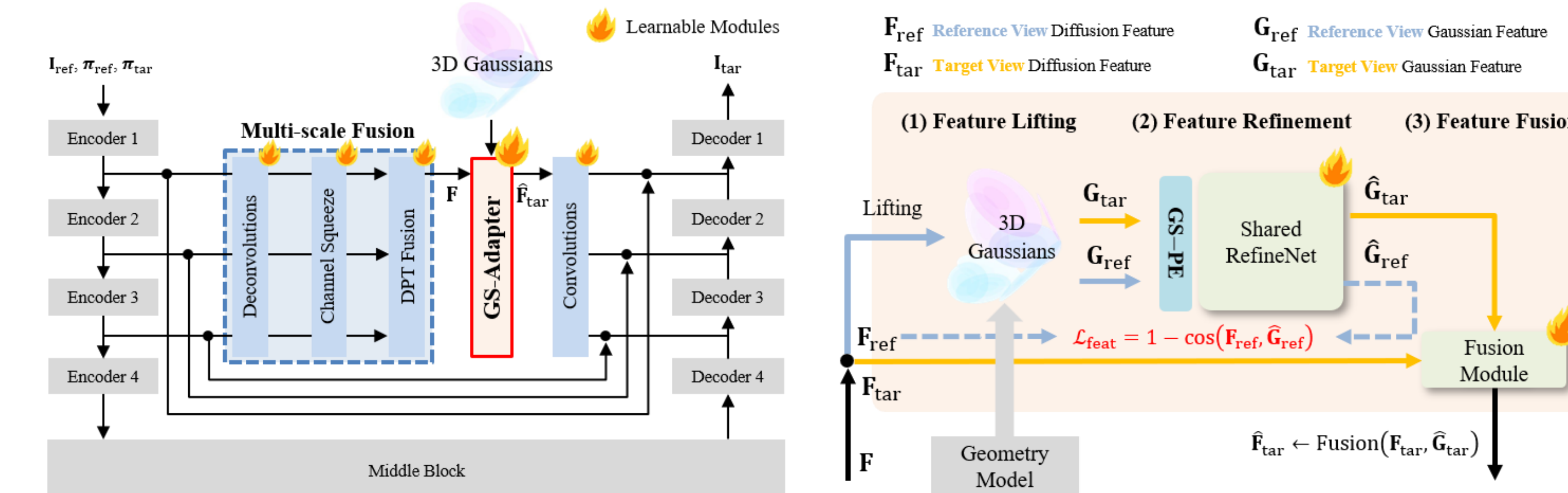


Our Method : Iterative Refinement with Geometry Prior

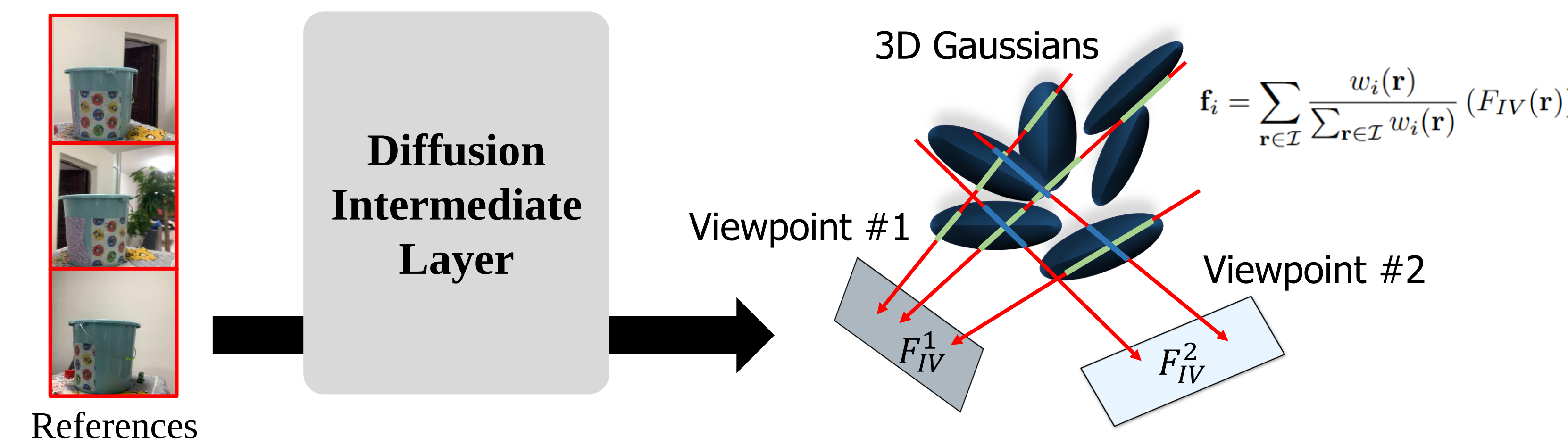


Method

● Overall Architecture and GS-Adapter

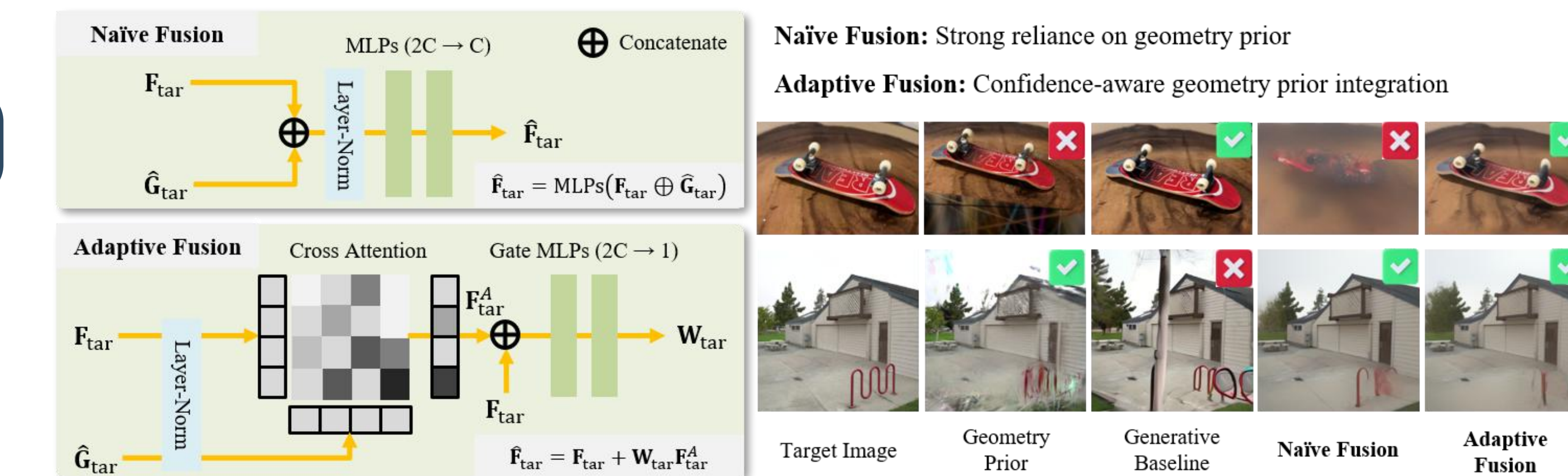


● Diffusion Feature Uplifting to 3D Gaussians

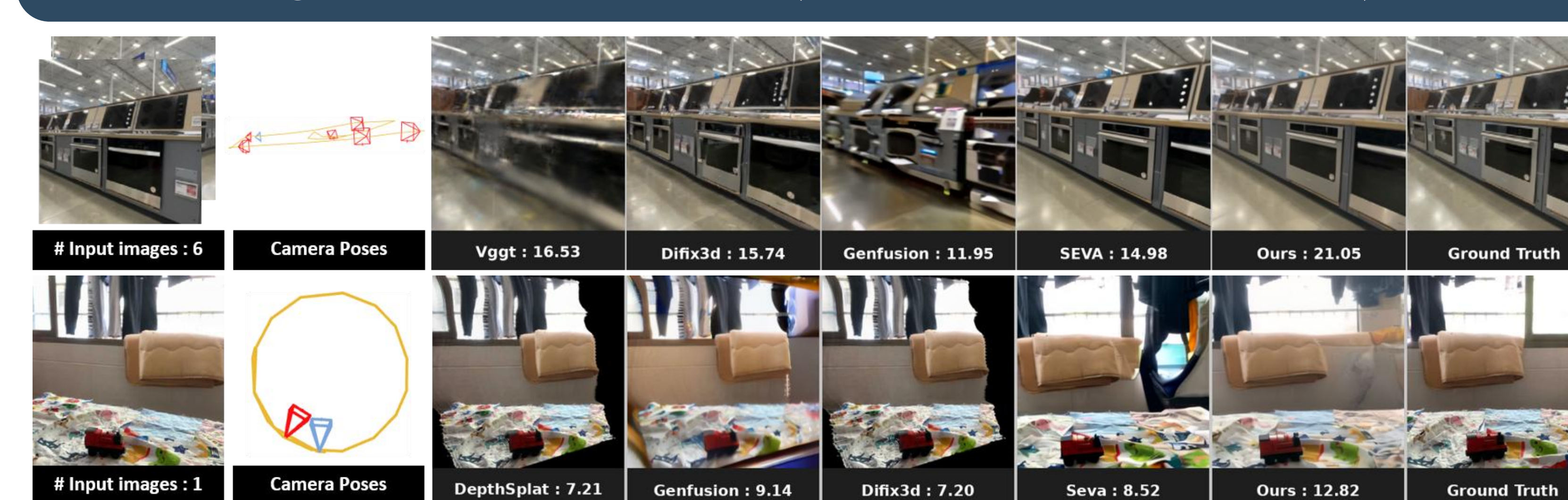


References

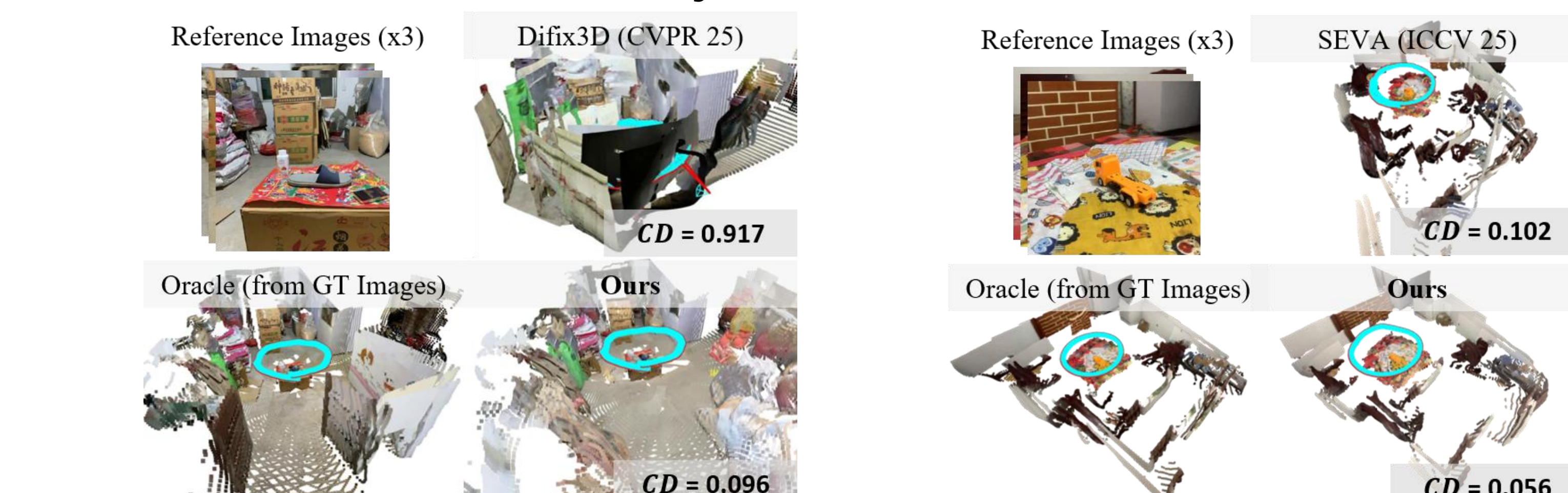
● Adaptive Fusion : Confidence-Guided Feature Fusion



Qualitative Results (PSNR / Chamfer Distance)



● Structure from Motion by Generated Video



Experiments

● Image Quality Metric for Novel View Synthesis

PSNR $\uparrow$ · bold = best · P = # input views															
<i>Method</i>	RE10K			DTU		CO3D		WRGBD		Mip360	DL3DV	T&T		Average	
<i>split / P</i>	D-1	R-1	R-3	R-1	R-3	V-1	R-3	S _c -3	S _n -6	R-6	S-6	L-9	O-2	O-3	
Feed-forward geometry models															
DepthSplat [CVPR 25]	13.02	14.49	26.99	8.79	14.19	11.29	15.08	13.92	13.50	15.04	15.87	16.88	15.76	17.99	15.67
VGGT [CVPR 25]	13.73	13.95	27.48	7.05	17.15	10.48	15.39	11.09	12.56	15.49	17.09	18.31	20.92	23.96	16.76
Generative models															
CameraCtrl [ICLR 25]	13.08	14.09	–	8.72	–	13.70	–	–	–	–	–	–	–	–	12.29
ViewCrafter [TPAMI 25]	14.37	17.44	15.83	9.22	8.49	15.87	9.89	9.23	9.37	10.81	10.51	9.75	10.74	11.13	12.02
SEVA [ICCV 25]	14.51	15.25	24.44	10.54	17.03	16.29	17.22	15.96	15.71	15.38	15.84	16.87	16.35	18.85	17.15
Generative model + Geometry prior															
MVSplat360 [NeurIPS 25]	13.68	13.44	20.97	9.10	11.38	12.69	14.29	14.34	13.22	14.15	15.09	15.26	14.22	15.84	14.86
GenFusion [CVPR 25]	14.87	15.88	22.69	8.36	13.75	12.15	12.99	13.38	12.47	11.99	13.11	14.78	18.25	20.14	15.30
Difix3D [CVPR 25]	12.92	14.37	27.14	8.79	17.22	11.93	15.23	12.60	12.67	14.95	16.33	17.33	20.68	23.36	16.89
Ours	16.66	17.62	27.11	12.12	19.18	17.14	19.26	18.23	17.12	16.57	18.11	19.66	18.76	22.24	19.09

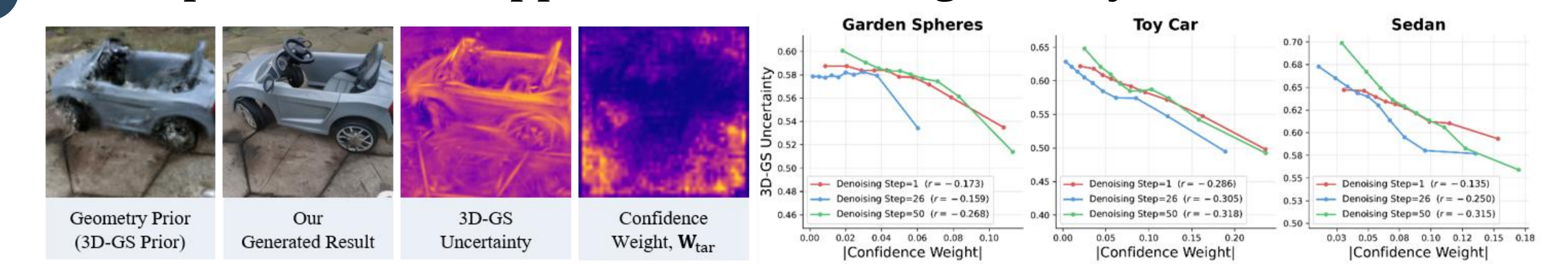
● Geometry Fidelity Metric for Novel View Synthesis

Method	Mip360 (3 ref. views)			DL3DV (6 ref. views)			WRGBD-S _n (3 ref. views)		
	T _{err} ↓	R _{err} ↓	CD ↓	T _{err} ↓	R _{err} ↓	CD ↓	T _{err} ↓	R _{err} ↓	CD ↓
Baselines									
SEVA [ICCV 25]	18.76	3.29	0.07	13.85	13.01	19.68	2.80	0.57	15.68
Generative model + VGGT [CVPR 25]									
Difix3D [CVPR 25]	219.39	51.64	0.68	12.92	11.82	64.4	9.66	3.76	16.11
SEVA + Input-level injection	233.03	99.79	1.02	13.14	12.03	81.14	17.19	3.82	16.76
Ours	16.24	2.98	0.07	15.11	14.07	13.27	2.01	0.41	18.11

● GS-Adapter transfers zero-shot to unseen geometry prior.



● Adaptive Fusion suppresses unreliable geometry.



Key Contributions

- Feature-level fusion of 3D Gaussians achieves robustness to rasterization noise in diffusion-based NVS.
- Adaptive gating selectively trusts geometry or diffusion features per pixel.
- GeoNVS achieves SOTA fidelity and geometric consistency across 8 benchmarks.

Project Page

